

Pregunta 1

R// Es 0 en $T=0K$

Pregunta 2

$$n_i^2 = B T^3 e^{-E_g/KT}$$

$$n = \frac{(1.7 \times 10^{10})^2}{2 \times 10^6} = 14450 \text{ cm}^{-3}$$

$$1.45 \times 10^4 \text{ cm}^{-3}$$

Pregunta 3

$$D_n = 36 \text{ cm}^2 \text{ s}^{-1}$$

$$d = 2 \mu\text{m}$$

$$n = 2.7 \times 10^6 \text{ cm}^{-3} \quad \text{a} \quad n = 10^{15} \text{ cm}^{-3}$$

$$J_n = q D_n \frac{dn}{dx}$$

$$J_n = (1.60 \times 10^{-19}) (36 \text{ cm}^2 \text{ s}^{-1}) \left(\frac{10^{15} - 2.7 \times 10^6}{2 \mu\text{m}} \right)$$

$$J_n = -748.8 \frac{\text{A}}{\text{cm}^2}$$

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Pregunta 4

$$J_n - \text{despl} = q \mu_e n \times E$$

$$E = \frac{J_n}{q \mu_e n}$$

$$E = = \frac{748.8}{(1.60 \times 10^{-19})(1350)(-2 \cdot 10^{15})}$$

$$E = 3466.66 \text{ V/cm}$$

Pregunta 5

$$LP = \sqrt{D_p \tau_p}$$

$$\tau_p = \frac{(0.02 \text{ cm})^2}{2.7 \text{ cm}^2/\text{s}} = 1.48 \times 10^{-5} \text{ s}$$

$$\tau_p = 14.8 \mu\text{s}$$

R// $N_A = 1 \times 10^{16} / \text{cm}^3$

$N_D = 1 \times 10^{18} / \text{cm}^3$

La concentración de elec y huecos en las reg tipo p y n en equi termico es

$n_p = 1.5 \times 10^{10} \text{ cm}^{-3}$

$p_n = 1.5 \times 10^{10} \text{ cm}^{-3}$

Pregunta 6

$N_p = N_A = 10^{16} \text{ cm}^{-3}$

Pregunta 7

$n_p = \frac{n_i^2}{N_A} = \frac{(1.5 \times 10^{10})^2}{1 \times 10^{16}}$

$N_p = 2.25 \times 10^4 \text{ cm}^{-3}$

Pregunta 8

$n_n = N_D = 10^{18} \text{ cm}^{-3}$

Pregunta 9

$p_n = \frac{n_i^2}{N_D} = \frac{(1.5 \times 10^{10})^2}{10^{18}} = 225 \text{ cm}^{-3}$

Pregunta 10 $V_{bi} = \frac{KT}{q} \ln \left(\frac{N_D N_A}{n_i^2} \right)$

$N_A N_D = n_i^2 e^{\frac{qV_{bi}}{KT}}$

$V_{bi} = (25 \times 10^{-3}) \ln \left(\frac{10^{16} \times 10^{18}}{(1.5 \times 10^{10})^2} \right) = 0.78 \text{ V}$

Pregunta 11

$$W = \sqrt{\frac{2(11.7)(8.8 \times 10^{-12})(0.817)}{1.6 \times 10^{-14}} \left(\frac{1}{10^6} + \frac{1}{10^{18}} \right)}$$

$$W = 3.27 \times 10^{-4}$$

$$W = 327 \mu\text{m}$$



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